

Exercise 06
Feasibility and Stability

Exercise 1 MPC and Feasible Set

Consider the discrete-time linear system

$$x(k+1) = Ax(k) + B(k) = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.5 \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k).$$

We define the constrained finite time optimal control problem

$$\begin{aligned} \min_U \quad & x_N^T P_N x_N + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i \\ \text{s.t.} \quad & x_{i+1} = Ax_i + Bu_i, \quad i \in \{0, \dots, N-1\} \\ & -1 \leq u_i \leq 1, \quad i \in \{0, \dots, N-1\} \\ & \begin{bmatrix} -15 \\ -15 \end{bmatrix} \leq x_i \leq \begin{bmatrix} 15 \\ 15 \end{bmatrix}, \quad i \in \{0, \dots, N-1\} \\ & x_N \in \mathcal{X}_f \\ & x_0 = x(k). \end{aligned}$$

1. If we set $\mathcal{X}_f = [0, 0]^T$, how would you choose P_N ? Why?
2. Assume $N = 5$, $\mathcal{X}_f = [0, 0]^T$, $Q = I$, $R = 0.1$, and $P = 0$. Consider a grid of initial conditions in the interval $-15 \leq x_1 \leq 15$, $-15 \leq x_2 \leq 15$, plot the grid point if the problem is feasible, otherwise not, and look at the size of the feasible set.
Hint: You can transform the optimal control problem to a QP using the approach with substitution as seen in Lec. 4 and then use the MATLAB function `quadprog` to solve the problem for different initial conditions and horizon lengths.
3. How does the feasible set change if we increase the horizon length to $N = 10$?

Exercise 2 Convexity and stability

Recall the definition of the Minkowski sum of two sets X and Y :

$$X \oplus Y := \{x + y \mid x \in X, y \in Y\}$$

1. Let X_1 and X_2 be convex invariant sets for the system $x(k+1) = Ax(k)$. Show that $\alpha X_1 \oplus (1-\alpha)X_2$ is also an invariant set for any $\alpha \in [0, 1]$.
2. Let $X_1 \subseteq \mathcal{X}$ and $X_2 \subseteq \mathcal{X}$, where X_1 , X_2 and $\mathcal{X}$ are convex sets. Show that $\alpha X_1 \oplus (1-\alpha)X_2 \subseteq \mathcal{X}$ for any $\alpha \in [0, 1]$.

3. Let $V_i(x) := x^T P_i x$ be a Lyapunov function for the system $x(k+1) = Ax(k)$ for $i = 1, 2$, with a rate of decrease of $x^T \Gamma x$, i.e.:

$$V_i(x(k+1)) - V_i(x) \leq -x^T \Gamma x.$$

Show that $V(x) = \alpha V_1(x) + (1 - \alpha)V_2(x)$ is also a Lyapunov function with a rate of decrease of $x^T \Gamma x$ for any $\alpha \in [0, 1]$.

4. Let K be a stabilizing controller for the system $x(k+1) = Ax(k) + Bu(k)$, and $X_i \subset \mathcal{X}$ be convex invariant sets for the system $x(k+1) = (A + BK)x(k)$, with $KX_i \subset \mathcal{U}$ for each $i = 1, 2$, where $\mathcal{U}$ is convex. $V_i(x) = x^T P_i x$, $i = 1, 2$, are Lyapunov functions for the system $x(k+1) = (A + BK)x(k)$ with a rate of decrease of $Q + K^T R K$, for some $Q = Q^T \succ 0$ and $R = R^T \succ 0$.

$$\begin{aligned} J^*(x(k)) = \min & \alpha V_1(x_N) + (1 - \alpha)V_2(x_N) + \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i \\ \text{s.t. } & x_{i+1} = Ax_i + Bu_i, \quad i = 0, \dots, N-1 \\ & x_i \in \mathcal{X}, \quad i = 0, \dots, N-1 \\ & u_i \in \mathcal{U}, \quad i = 0, \dots, N-1 \\ & x_N \in \alpha X_1 \oplus (1 - \alpha)X_2 \\ & x_0 = x(k) \end{aligned}$$

Prove that this MPC controller is stabilizing and recursively feasible for any $\alpha \in [0, 1]$ by

- listing sufficient conditions for stability and
- proving them

Hint: You can use the results of the previous three questions, even if you couldn't answer them.

Exercise 3 Invariant Terminal Set

Consider the following standard MPC optimization problem:

$$\begin{aligned} J^*(x(k)) = \min & x_4^T P x_4 + \sum_{i=0}^3 x_i^T x_i + u_i^T u_i \\ \text{s.t. } & x_{i+1} = Ax_i + Bu_i, \quad i = 0, \dots, 3 \\ & x_i \in \mathcal{X}, \quad i = 1, \dots, 3 \\ & u_i \in \mathcal{U}, \quad i = 0, \dots, 3 \\ & x_4 \in \mathcal{X}_f \\ & x_0 = x(k), \end{aligned}$$

where P is the solution of the Algebraic Riccati equation, K is the infinite horizon LQR controller, and $\mathcal{X}_f$ is the maximal invariant set for the closed-loop system $x(k+1) = (A + BK)x(k)$.

At time $k = 0$, the system is in state $x(0)$, and the optimal solution to this problem is given by the state and input sequences $\{x_0^*, x_1^*, \dots, x_4^*\}$ and $\{u_0^*, \dots, u_3^*\}$.

At time $k = 0$, we apply the optimal control input $u(0) = u_0^*$ and the system moves to state $x(1) = Ax(0) + Bu(0)$.

1. Given the information you have above, design a control sequence that does not require solving an optimization problem, and will take our system from its current state and cause it to converge to the origin, satisfying all constraints on the way *without solving any optimization problems*. Give the first 5 steps of this sequence $\{u(1), u(2), \dots, u(5)\}$.
2. If you can write down such a controller given only the solution at time $k = 0$, why do we bother to re-optimize every time-step in MPC? Give two reasons.

Exercise 4 **Stability and Recursive Feasibility of Nominal MPC**

Consider the scalar system

$$x(k+1) = 2x(k) + u(k)$$

and the MPC control law $\kappa(x(k)) = u_0^*(x(k))$ defined by

$$\begin{aligned} P_1 : \underset{X, U}{\text{minimize}} \quad & \sum_{i=0}^{N-1} qx_i^2 + ru_i^2 \\ \text{subject to} \quad & x_{i+1} = 2x_i + u_i, \quad i = 0, \dots, N-1, \\ & u_i \in [-2, 2], \quad i = 0, \dots, N-1, \\ & x_N \in [-1, 1], \\ & x_0 = x(k) \end{aligned}$$

- i) Compute the gain K_∞ of the optimal controller $\kappa_\infty(x(k)) = K_\infty x(k)$ for the *unconstrained* infinite-horizon optimal control problem, i.e., for $N \rightarrow \infty$, with $q = r = 1$.
- ii) Show that the set $[-1, 1]$ is invariant under $\kappa_\infty(x(k))$ and that input constraints are satisfied, that is $\kappa_\infty(x(k)) \in [-2, 2]$ for all $x(k) \in [-1, 1]$.
- iii) Is the MPC problem P_1 recursively feasible? Explain your reasoning.
- iv) Is the closed-loop system under the MPC control law $\kappa(x(k))$ guaranteed to stay within a bounded set around the origin, given the MPC problem is feasible at the initial state? Is it guaranteed to be asymptotically stable? If not, propose a modification to P_1 such that the closed-loop system is asymptotically stable. Explain your reasoning.